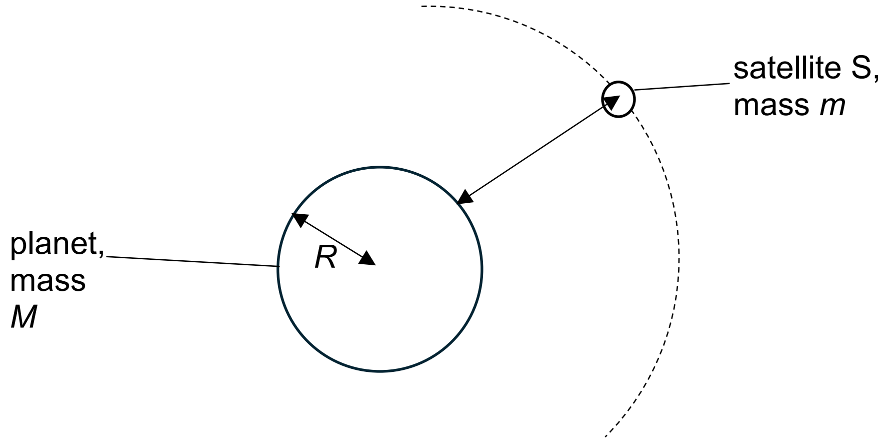
2	**(a)**	A satellite S of mass m is in a stable circular orbit at an altitude of $2R$ above the surface of a planet of mass M and radius R, as shown in Fig. 2.1. Assume the planet has no atmosphere and that all its mass is concentrated at its centre.  **Fig. 2.1**	
	(a)	Show that the kinetic energy E_k of the satellite S in orbit is given by the expression: $$E_k = \frac{GMm}{6R}$$ where G is the gravitational constant.	
			[2]
		Solution: The gravitational force provides the centripetal force. $F_G = F_C$ $\frac{GMm}{(3R)^2} = \frac{mv^2}{3R}$ $v^2 = \frac{(3R)Mm}{(3R)^2 m} = \frac{GM}{3R}$ $KE = \frac{1}{2}mv^2 = \frac{1}{2}m\frac{GM}{3R}$ $= \frac{GMm}{6R}$	B1 B1

(b) The planet has mass 4.5×10^{24} kg and radius of 5.5×10^{3} km. The satellite has a mass of 1500 kg.

Determine the total energy of satellite S in orbit.

total energy = ………………………………. J [2]

Solution:

Total energy = $-\frac{GMm}{2r}$ where r is the orbital radius and is equal to 3R for this Q

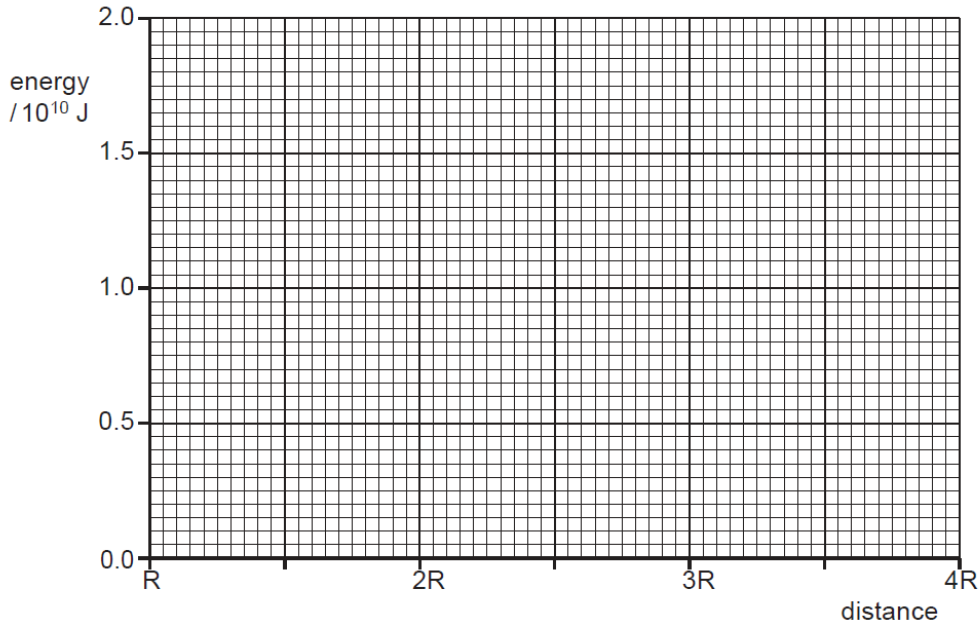
$$= \frac{(6.67 \times 10^{-11})(4.5 \times 10^{24})1500}{2 \text{ x } 3(5.5 \times 10^{3} \text{x} 10^{3})}$$ M1

= $- 1.36 \times 10^{10}$ J (negative sign to be included) A1

If students equate E_k to E_T, there must be some derivation shown to get M1 marks.

(c) A second satellite P is launched into orbits from the surface of the same planet with an initial kinetic energy of 2.0×10^{10} J. It rises to a distance of 4R from the centre of the planet.

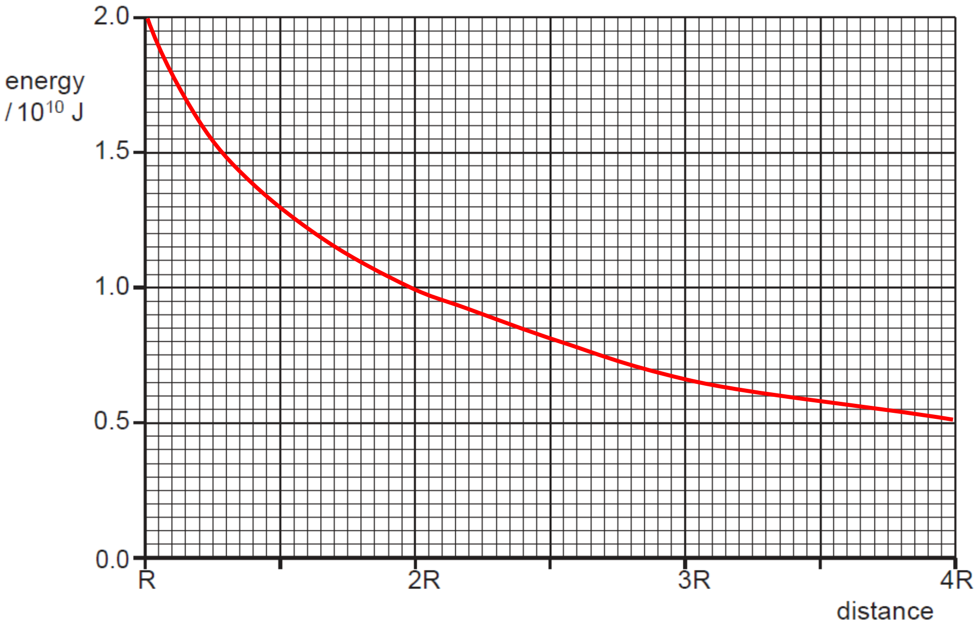
On the axes provided in Fig 2.2, sketch a graph to show how the satellite's orbital kinetic energy varies with distance from the centre of the planet as it moves from R to 4R.



[2]

Fig 2.2

Solution:



B1 – Shape of graph:

- Smooth curve decreasing with increasing r (concave upwards)

1 mark – Correct key points labelled or plotted:

- Correctly labels or plots the following values:

Since $E_k \ \alpha \ \frac{1}{r}$

r	E_K / 10^{10} J
R	+2.0
2R	+1.0
3R	+0.67
4R	+0.5

(d) A third satellite Q is to be launched vertically from the surface of the same planet. Determine the minimum speed that satellite Q must be given at the surface to escape the planet's gravitational field.

minimum speed = ... m s^{-1} [2]

		Solution: By conservation of energy, Energy of satellite Q at the surface of the planet = energy of satellite Q at infinity $$\frac{1}{2}mv^2+\left(\frac{-GMm}{r}\right)=0$$ $$v=\sqrt{\frac{2GM}{r}}=\sqrt{\frac{2(6.67\times10^{-11})(4.5\times10^{24})}{5.5\times10^{3}\text{x}10^{3}}}$$ $v = 1.04 \times 10^4$ m s^{-1}	M1 A1

[Total: 8]